CASE-STUDY PRACTICE | CAPSTONE 3 | MOCK INTERVIEW EDITION

Mock Interview Edition

Video Recommendation: Retrieve Cheaply, Rank Carefully

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

Contents

Every line below is a link: click it to jump to that page. The outline panel of your PDF viewer lists the same entries, and the “Contents” link in every page footer brings you back here. Question numbers are the same as in the Study Edition.

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Every scenario, by part

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03 Does a watch-time weight favor long videos? Risk 7

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

Round 1 Objective and sizing 4

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02 What does the weighted loss learn? Calculation 5
03 Does a watch-time weight favor long videos? Risk 7
04 Size the funnel Estimation 8
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15 Where does the ranking CPU go? Cost 25
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05 A random held-out watch, or the next watch? Labels 9
09 What does a sampled softmax save? Training cost 16
13 Retrieve from 10 million videos in tens of milliseconds Serving 22
15 Where does the ranking CPU go? Cost 25

SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: the objective, the labels and the funnel

Choosing what to optimize, reading what a weighted loss actually learns, and sizing the funnel from the corpus to the screen.

01 Rank by click, or by watch time?

OBJECTIVE

SCENARIO

A product manager wants the ranker to maximize click-through rate, since clicks are easy to count. The paper says that ranking by click-through rate often promotes deceptive videos that users do not complete, and that watch time better captures engagement. Make the case with numbers.

OPENING PROBE

What does each metric say about two videos?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the click metric rewarding?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How does the paper change the objective?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you check before adopting it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 What does the weighted loss learn?

CALCULATION

SCENARIO

The paper weights each clicked impression by its watch time T and each unclicked one by 1, then uses the exponential of the model's raw score at inference. An interviewer asks you to show why that score estimates expected watch time. Use 1,000 impressions with 20 clicks, each clicked impression watched for 5 minutes.

OPENING PROBE

What does weighted logistic regression learn?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Compute it for the example.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is the approximation?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

When does the approximation weaken?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

Why the exponential at inference?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Does a watch-time weight favor long videos?

RISK

SCENARIO

A reviewer notes that a watch-time weight gives a clicked impression of a 60-minute video that is watched to the end a weight of 3,600 seconds, and a 3-minute video watched to the end 180. The paper does not discuss this. Is it a problem, and what would you do?

OPENING PROBE

How large is the difference?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the paper say?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why might length bias still matter?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you measure and try?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Size the funnel

ESTIMATION

SCENARIO

The paper's Figure 2 shows the funnel from millions of videos to hundreds of candidates to dozens shown, and says only that. A product owner asks how many candidates to keep, and what that does to cost. Use 1 billion users, 2 page loads each a day, 10 million videos, 500 candidates and 20 shown.

OPENING PROBE

What are the ratios and the rate?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Which stage does the candidate count affect?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does doubling the candidates cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is the ceiling?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Data and labels: examples, leakage and freshness

Which watches to train on, how to build a label that does not leak the future, what to hide from the model, and how to tell it about time.

05 A random held-out watch, or the next watch?

LABELS

SCENARIO

A colleague builds training labels the way many collaborative filtering systems do: remove one watch at random from a user's history and predict it from the others. The paper says predicting the next watch worked much better in live tests. Explain why, using a user who watches a series in order.

OPENING PROBE

What does the random holdout do to the series?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why can't that work at serving time?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the rollback do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why is this hard to see in offline tests?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 The model that copies the last search

LABELS

SCENARIO

A user has just searched for a singer, and the model must predict the next video. A first version scores well offline and then performs poorly on the homepage. The paper gives this exact example. Find the cause and fix it.

OPENING PROBE

What does the model learn?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Why does it look good offline?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is the fix?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Would a bigger model help?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What should you review in other features?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Teach the model what time it is

FRESHNESS

SCENARIO

Fresh uploads are important to the product, but a model trained on weeks of history underrates a video uploaded yesterday. The paper feeds the age of each example as a feature and sets it to zero at serving. Explain the bias, the fix and the trap.

OPENING PROBE

What is the bias?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does example age do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Why zero at serving?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What is the trap?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does it not fix?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the example builder

CODE REVIEW

SCENARIO

A colleague writes the function below to build training examples from a user's watch history, and the training loss looks excellent. In a live test the recommendations are poor, and heavy users seem to be served much better than light ones. Find the faults.

```
def build_examples(history):
    # history: list of (timestamp, video_id), oldest first
    examples = []
    for i, (t, label) in enumerate(history):
        context = [v for j, (_, v) in enumerate(history) if j != i]
        examples.append((context, label))
    return examples
```

OPENING PROBE

What is wrong with the context?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is wrong with the count of examples?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What else is wrong?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

Candidate generation: classification among millions and fast retrieval

Why the first stage is a classifier, what a sampled softmax saves, what the network holds, and how to retrieve from millions of videos in tens of milliseconds.

09 What does a sampled softmax save?

TRAINING COST

SCENARIO

Candidate generation predicts the next watch among millions of videos, so an ordinary softmax sums over every video for every training example. The paper samples several thousand negatives instead and reports a speedup of more than 100 times. Check that claim with your own arithmetic.

OPENING PROBE

What does the full softmax cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the sampled version cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the paper actually say?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does sampling change in the math?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

10 Why not a hierarchical softmax?

DESIGN CHOICE

SCENARIO

A teammate points out that a hierarchical softmax is much cheaper than sampling: it needs only about 20 binary decisions for a million classes. The paper tried it and went with sampling. Explain why a cheaper method lost.

OPENING PROBE

How cheap is the tree?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

So why did the paper not use it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the stated reason?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you do if you had to decide yourself?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

11 Where do the parameters live?

MODEL DESIGN

SCENARIO

The candidate generation network averages embeddings of watched videos and search tokens, concatenates demographics, and feeds ReLU layers. A manager asks what to cut if the model must be smaller. Use the paper's experimental configuration.

OPENING PROBE

What is the architecture?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

How many parameters are in the embeddings?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

How many are in the layers?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does depth buy?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you cut?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

12 Review the retrieval function

CODE REVIEW

SCENARIO

A colleague writes the retrieval step below and tests it on 10,000 videos, where it works. In staging, with 10 million videos, some requests return garbage and the rest are slow. The paper says calibrated probabilities are not needed at serving time. Find the faults.

```
import numpy as np

def generate_candidates(u, video_vecs, video_ids, top_n=500):
    scores = video_vecs @ u
    probs = np.exp(scores) / np.exp(scores).sum()    # softmax over the corpus
    order = np.argsort(-probs)
    return [video_ids[i] for i in order[:top_n]]
```

OPENING PROBE

What does the softmax add?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What goes wrong with it at scale?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is wrong with the sort?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Is the fixed version enough?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix, and the test.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

13 Retrieve from 10 million videos in tens of milliseconds

SERVING

SCENARIO

The paper says scoring millions of items in tens of milliseconds requires an approximate scheme that is sublinear in the number of classes, and that A/B results were not sensitive to the choice of algorithm. You have 10 million videos of 256 float32 values and 23,148 requests a second. Decide how to retrieve.

OPENING PROBE

What does an exact scan cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Does batching help?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does an approximate index do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does it risk?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

PART 4

Ranking, evaluation and operations

Features and shared embeddings, where ranking cost sits, fresh signals and churn, reading offline and online evidence, and scaling traffic.

14 Share the video embedding across features

FEATURES

SCENARIO

The ranking model uses hundreds of features. Several of them are video IDs: the impression's video, the last video the user watched, and the video that seeded the recommendation. An engineer wants a separate embedding table for each, for flexibility. Decide, using the paper and the parameter arithmetic.

OPENING PROBE

How big is one embedding table?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What would separate tables cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does sharing give?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

How do features stay distinct?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What other feature details matter?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

15 Where does the ranking CPU go?

COST

SCENARIO

Ranking CPU is the paper's binding constraint: its best tower is the largest one that stays within the serving CPU budget. A manager asks whether dropping the third layer is worth it. Use the Step 2 figures and the paper's Table 1.

OPENING PROBE

How does the work split?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the table say about the last layer?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does dropping it save?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What other levers are there?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

16 Fresh signals and churn

OPERATIONS

SCENARIO

A feature such as how many times this video was already shown to this user is computed from the last few seconds of activity. A platform engineer proposes refreshing these features once a day, to save infrastructure cost. Say what happens.

OPENING PROBE

What does the feature do in the paper?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What happens with a daily refresh?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What else is lost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What is the cost of doing it right?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you decide?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

17 Read the offline numbers, then decide

EVALUATION

SCENARIO

The paper reports a weighted, per-user loss falling from 41.6% to 34.6% as hidden layers are added, and says that live A/B results are not always correlated with offline ones. A teammate wants to ship the best offline model without a live test. Explain what the numbers say and what they cannot.

OPENING PROBE

How is the metric computed?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does Table 1 show?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What do the numbers share with the model?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does the paper say about offline against online?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you ship on?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

18 Ten times the traffic

SCALE

SCENARIO

Imagine ten times the users or the page loads. A manager asks what breaks first, and what to add. Use the Step 2 anchors.

OPENING PROBE

What happens to the rate?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What happens to ranking?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What happens to retrieval?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does not follow requests?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Rank by click, or by watch time?	5			
02	What does the weighted loss learn?	6			
03	Does a watch-time weight favor long videos?	5			
04	Size the funnel	5			
05	A random held-out watch, or the next watch?	5			
06	The model that copies the last search	6			
07	Teach the model what time it is	6			
08	Review the example builder	6			
09	What does a sampled softmax save?	5			
10	Why not a hierarchical softmax?	5			
11	Where do the parameters live?	6			
12	Review the retrieval function	6			
13	Retrieve from 10 million videos in tens of millise...	5			
14	Share the video embedding across features	6			
15	Where does the ranking CPU go?	5			
16	Fresh signals and churn	6			
17	Read the offline numbers, then decide	6			
18	Ten times the traffic	5			